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**Department of Electronics and Communication
Engineering**

Automotive Electronics Course Project Report

**RFID Based Ignition System
(Semester VI)**

Course Code: 25EECC305

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Academic Year: 2025 - 2026



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CERTIFICATE

This is to certify that the project entitled **RFID Based Ignition System** is a bonafide work carried out by **Parth Jadhav [USN: 02FE23BEC083]** **Md. Danish Shamsheer [USN: 02FE23BEC112]** and **Roopashree Pote[USN: 02FE23BEC122]** students of VI semester of KLE Technological University, Dr. M. S. Sheshgiri Campus, Belagavi, in partial fulfillment for the award of the degree of **Bachelor of Engineering** in the Department of **Electronics and Communication Engineering** during the academic year **2025–2026**. It is certified that all corrections suggested during internal assessment have been incorporated in the final report.

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DECLARATION

We hereby declare that the Automotive Electronics Course Project presented in this report, entitled “**RFID Based Ignition System**” submitted to KLE Technological University for the completion of the Automotive Electronics Project, is the original work carried out by us in the Department of Electronics and Communication Engineering, KLE Technological University, Dr. M. S. Sheshgiri Campus, Belagavi, under the guidance of **Dr. Vijay Rayar**, Department of Electronics and Communication Engineering.

We further declare that, to the best of our knowledge and belief, the work reported herein has not been submitted as part of any other project or for the award of any course, degree, or diploma at this or any other university or institution. The results presented in this report are solely the outcome of our efforts.

We also confirm that all the work documented in this report has been completed by us.

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Acknowledgment

We sincerely express our gratitude to our guide, **Dr.Vijay Rayar**, for his constant support, encouragement, and guidance throughout the course of this project. His expert advice, constructive feedback, and insightful suggestions have been invaluable in shaping the direction of our work and ensuring its successful completion. His dedication and patience inspired us to overcome challenges and strive for excellence.

We would also like to acknowledge his efforts in providing us with the necessary resources and creating an environment conducive to learning and research. His motivation and constant encouragement kept us focused and confident throughout the project. We are truly thankful for his mentorship and for sharing his knowledge and experience so generously, which has greatly enriched our understanding and skills. We extend our sincere gratitude to the Head of the Department, **Dr.D.A.Torse**, for providing a conducive learning environment and fostering a culture of academic excellence.

Lastly, we express our heartfelt thanks to our Principal, **Dr.S.F.Patil**, for their unwavering support and encouragement. Your belief in our abilities has been a driving force behind our efforts. This project would not have been possible without the collective guidance and support of these esteemed individuals.

Abstract

The RFID Based Smart Ignition System is an intelligent automotive security system developed using Raspberry Pi Pico, RC522 RFID module, and L298N motor driver. The system allows only authorized RFID tags to start the ignition motor, thereby improving vehicle security and preventing unauthorized access. When a valid RFID tag is scanned, the Raspberry Pi Pico authenticates the UID and activates the motor along with the onboard LED indication.

A manual stop switch is also provided to stop the motor whenever required. Unauthorized RFID tags are denied access and the motor remains OFF. The proposed system is low-cost, reliable, and easy to implement for automotive security applications. The project demonstrates the practical application of RFID technology, embedded systems, and automotive electronics in smart vehicle ignition systems.

Contents

1	Introduction	9
2	Literature Survey	10
2.1	RFID Based Vehicle Security System	10
2.2	Smart Vehicle Ignition Using Embedded Systems	10
2.3	RFID Based Access Control System	11
2.4	IoT Based Vehicle Monitoring and Security System	11
3	Problem Statement and Objectives	12
3.1	Problem Statement	12
3.2	Objectives	12
4	System Requirements and Specifications	13
4.1	Hardware Requirements	13
4.2	Software Requirements	13
4.3	Specifications	14
5	Tools Used	15
5.1	MATLAB Simulink	15
5.2	Arduino IDE	15
5.3	MFRC522 RFID Library	15
5.4	Raspberry Pi Pico RP2040 Board Package	15
6	System Architecture, Block Diagram and Flow Chart	16
6.1	System Architecture	16
6.2	Block Diagram	17
6.3	Flow Chart	18
7	Model-Based Design	19
7.1	Simulation and Analysis	19
7.2	Software Components Used	20
7.3	Software Implementation	20
8	Hardware Implementation	21

8.1 Hardware Components Used.....	21
8.2 Raspberry Pi Pico Pin Connections.....	22
8.3 Hardware Implementation	22
9 Code	24
9.1 Arduino Code.....	24
10 Applications and Future Scope	27
10.1 Applications	27
10.2 Future Scope.....	27
11 Conclusion	28

Chapter 1

Introduction

Modern vehicles increasingly require intelligent security systems to prevent unauthorized access and improve user safety. Conventional key-based ignition systems can be duplicated or misused easily, leading to security risks and vehicle theft. RFID technology provides a secure and efficient solution for authentication and ignition control in automotive applications.

The proposed RFID Based Smart Ignition System uses an RC522 RFID module and Raspberry Pi Pico controller to authenticate RFID tags before allowing ignition access. The RFID reader scans the tag UID and sends it to the Raspberry Pi Pico for verification. If the scanned UID matches the authorized UID stored in memory, the system activates the ignition motor through the L298N motor driver and turns ON the onboard LED as a status indication.

A manual stop switch is also provided to stop the motor whenever required. Unauthorized RFID tags are denied access and the motor remains OFF. The proposed system is simple, reliable, low-cost, and suitable for modern automotive security applications. The project demonstrates the practical implementation of RFID technology, embedded systems, and automotive electronics in smart vehicle ignition systems.

Chapter 2

Literature Survey

2.1 RFID Based Vehicle Security System

In the paper titled “RFID Based Vehicle Security System”, the authors proposed a smart vehicle authentication system using RFID technology to improve vehicle security and prevent unauthorized access. The system uses RFID tags and readers to authenticate users before allowing vehicle ignition. The RFID reader scans the tag and sends the UID to the embedded controller for verification. If the scanned UID matches the authorized UID stored in memory, the ignition system is activated. The proposed system demonstrated reliable authentication, fast response, and improved vehicle protection. This work highlights the importance of RFID technology in modern automotive security systems and provides valuable concepts for developing secure ignition control applications.

2.2 Smart Vehicle Ignition Using Embedded Systems

In the paper titled “Smart Vehicle Ignition Using Embedded Systems”, the authors developed an intelligent ignition control system using embedded technology and electronic authentication methods. The primary objective of the system is to replace conventional mechanical keys with secure electronic ignition control. The system uses a microcontroller to process authentication signals and control vehicle ignition. The implementation demonstrated secure operation, reduced chances of vehicle theft, and improved reliability. The embedded system approach provided advantages such as low cost, fast response, compact design, and easy implementation. This paper provides useful knowledge for developing smart ignition systems using embedded controllers and automotive electronics.

2.3 RFID Based Access Control System

In the paper titled “RFID Based Access Control System”, the authors implemented an RFID authentication system using microcontrollers and RFID readers for secure access control applications. The system allows access only to authorized users and denies unauthorized entries. The RFID reader scans the RFID tag and sends the UID information to the controller for verification. If authentication is successful, the output device is activated; otherwise, access is denied. The proposed system demonstrated reliable performance, fast operation, and improved security. This work explains the practical implementation of RFID technology and serves as an important reference for developing secure automotive ignition systems.

2.4 IoT Based Vehicle Monitoring and Security System

In the paper titled “IoT Based Vehicle Monitoring and Security System”, the authors proposed a smart automotive security system using embedded technology and IoT communication. The system provides vehicle monitoring, authentication, and security control through intelligent embedded hardware. Sensors and controllers continuously monitor vehicle conditions and improve overall security and user convenience. The implementation demonstrated efficient operation, real-time monitoring, and secure control mechanisms. The paper highlights the growing importance of IoT and embedded systems in modern automotive applications and provides future enhancement possibilities for smart ignition and vehicle security systems.

Chapter 3

Problem Statement and Objectives

3.1 Problem Statement

Traditional vehicle ignition systems use mechanical keys which may be duplicated or accessed by unauthorized users. This creates security risks and may lead to vehicle theft. Conventional ignition systems also lack intelligent authentication and secure access control mechanisms.

Therefore, an RFID-based smart ignition system is required to improve vehicle security, provide authorized access control, and prevent unauthorized vehicle operation.

3.2 Objectives

- To develop an RFID-based smart ignition system
- To authenticate authorized RFID tags
- To control ignition motor using Raspberry Pi Pico
- To prevent unauthorized vehicle access
- To provide manual motor stop operation
- To understand automotive electronics concepts

Chapter 4

System Requirements and Specifications

4.1 Hardware Requirements

- Raspberry Pi Pico H
- RC522 RFID Module
- RFID Tags
- L298N Motor Driver
- DC Motor
- Push Button Switch
- 12V Adapter
- Breadboard
- Jumper Wires
- USB Cable

4.2 Software Requirements

- Arduino IDE
- MFRC522 RFID Library
- Raspberry Pi Pico RP2040 Board Package

4.3 Specifications

- RFID-based ignition authentication
- Authorized access control
- Motor ignition control
- Manual stop switch operation
- Real-time response
- Embedded automotive security system

Chapter 5

Tools Used

5.1 MATLAB Simulink

MATLAB Simulink was used to design and simulate the RFID Based Smart Ignition System before hardware implementation. The simulation was carried out using logical blocks, switches, and control signals to verify the working of the ignition system.

5.2 Arduino IDE

Arduino IDE was used for writing, compiling, and uploading the program into the Raspberry Pi Pico controller. It provided a simple platform for implementing RFID authentication and motor control logic.

5.3 MFRC522 RFID Library

The MFRC522 RFID library was used for interfacing the RC522 RFID module with the Raspberry Pi Pico. The library enabled RFID tag detection, UID reading, and authentication operations.

5.4 Raspberry Pi Pico RP2040 Board Package

The Raspberry Pi Pico RP2040 board package was used to program the Raspberry Pi Pico using Arduino IDE. It provided support for GPIO, SPI communication, and onboard peripheral control required for the project.

Chapter 6

System Architecture, Block Diagram and Flow Chart

6.1 System Architecture

The system architecture consists of RFID input section, controller section, motor driver section, and output section.

- RC522 RFID module is used to read RFID tags
- Raspberry Pi Pico processes the RFID UID
- Authorized RFID tag activates the ignition motor
- L298N motor driver controls the DC motor
- Stop switch is used to stop the motor manually
- Onboard LED acts as the status indication output

6.2 Block Diagram

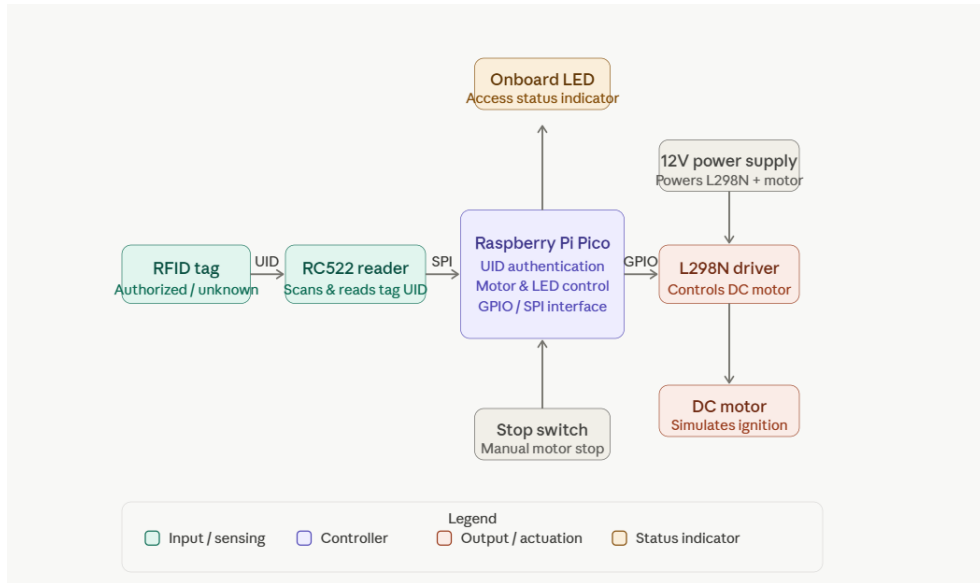


Figure 6.1: Block Diagram of RFID Ignition System

The block diagram illustrates the operation of the RFID Based Smart Ignition System. The RFID tag is scanned using the RC522 RFID reader module, which sends the UID information to the Raspberry Pi Pico controller.

The Raspberry Pi Pico compares the scanned UID with the authorized UID stored in memory. If the UID matches, the controller activates the onboard LED and enables the L298N motor driver to start the ignition motor. If the scanned RFID tag is unauthorized, the controller denies access and keeps the motor OFF.

A stop switch is connected to the Raspberry Pi Pico and is used to stop the motor manually whenever required. The system provides secure authentication, reliable operation, and intelligent ignition control for automotive applications.

6.3 Flow Chart

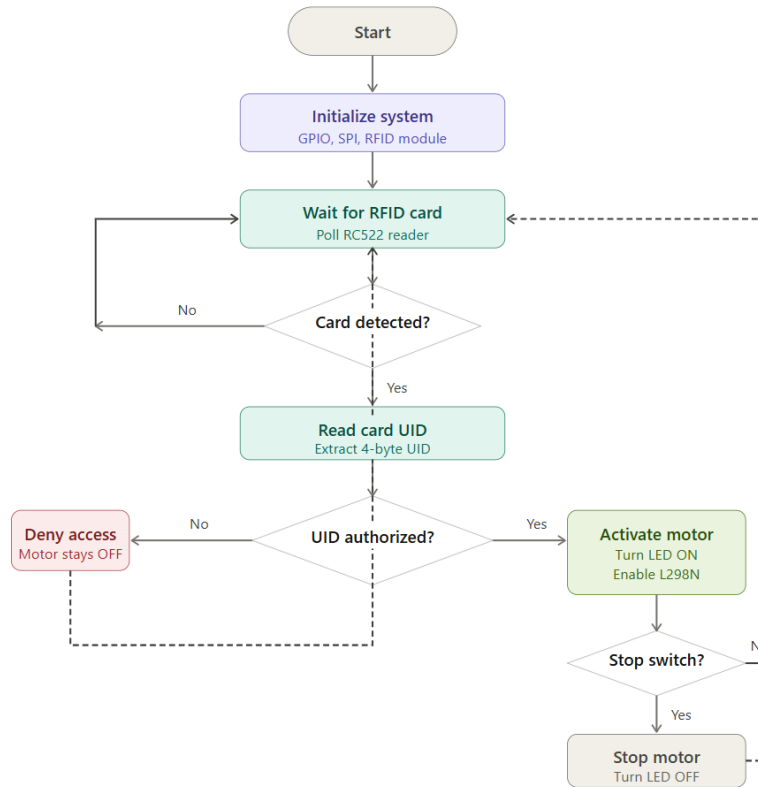


Figure 6.2: Flow Chart

The flowchart explains the complete working of the Interior Light Control System with day and night detection.

Initially, the STM32 controller initializes all input and output pins. The system reads the night sensor to identify whether the surroundings are dark or bright.

During daytime, automatic lighting is disabled and only the manual switch controls the LED.

During nighttime, the STM32 checks both door switch and manual switch conditions. If the door opens or manual switch is ON, the LED turns ON automatically.

The system continuously monitors all inputs and updates the LED output in real time.

Model-Based Design

7.1 Simulation and Analysis

The RFID Based Smart Ignition System was simulated using MATLAB Simulink before hardware implementation. The simulation verified RFID authentication, motor control logic, and stop switch operation.

When an authorized RFID signal is detected, the motor and onboard LED turn ON. If an unauthorized RFID tag is detected or the stop switch is pressed, the motor turns OFF immediately. The simulation helped in verifying the overall working of the ignition system before hardware implementation.

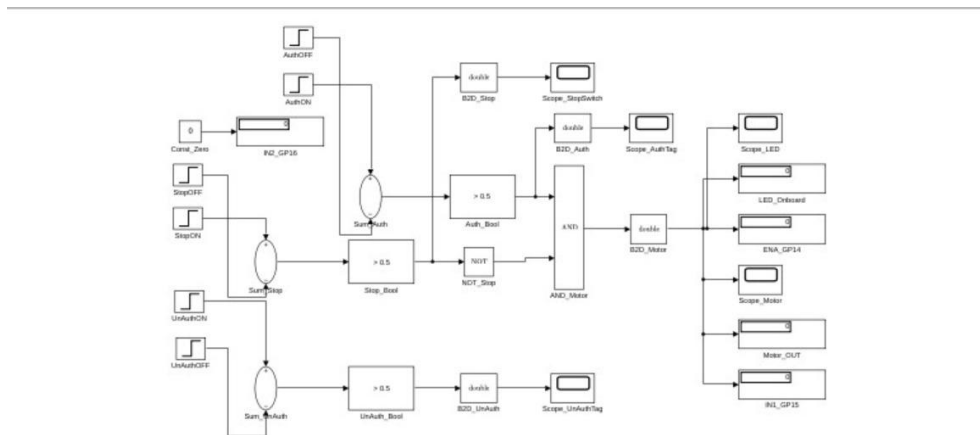


Figure 7.1: Simulation Model of RFID Based Smart Ignition System

7.2 Software Components Used

Sl. No	Block Name	Purpose
1	Constant Block	RFID input signal
2	Switch Blocks	Authorized/Unauthorized RFID
3	Stop Switch Block	Motor stop operation
4	Sum Blocks	Combines signals
5	Comparator Blocks	Authentication checking
6	NOT Gate	Inverts stop condition
7	AND Gate	Motor enable logic
8	Display/Scope Blocks	Displays output

Table 7.1: Software Components Used in Simulink Model

7.3 Software Implementation

The software implementation was carried out using MATLAB Simulink to verify the RFID authentication and motor control logic. Logical blocks were used to process RFID signals and stop switch conditions. Based on the input conditions, the system controls the motor and onboard LED outputs successfully.

Chapter 8

Hardware Implementation

8.1 Hardware Components Used

Sl. No	Component	Quantity
1	Raspberry Pi Pico H	1
2	RC522 RFID Module	1
3	RFID Tags	2
4	L298N Motor Driver	1
5	DC Motor	1
6	Push Button Switch	1
7	12V Adapter	1
8	Breadboard	1
9	Jumper Wires	10–15

8.2 Raspberry Pi Pico Pin Connections

Pico Pin	Connected Device	Function
GP2	RC522 SCK	SPI Clock Signal
GP3	RC522 MOSI	SPI Data Transmission
GP4	RC522 MISO	SPI Data Reception
GP5	RC522 SDA	RFID Module Select
GP6	RC522 RST	RFID Reset Signal
GP14	L298N ENA	Motor Enable Signal
GP15	L298N IN1	Motor Direction Control
GP16	L298N IN2	Motor Direction Control
GP17	Push Button Switch	Motor Stop Input
3.3V	RC522 VCC	Power Supply
GND	Common Ground	Ground Connection

Table 8.1: Raspberry Pi Pico Pin Connections

8.3 Hardware Implementation

The hardware implementation of the RFID Based Smart Ignition System was carried out using Raspberry Pi Pico, RC522 RFID module, L298N motor driver, DC motor, and stop switch. The RC522 RFID reader scans RFID tags and sends the UID information to the Raspberry Pi Pico controller through SPI communication.

The Raspberry Pi Pico compares the scanned UID with the authorized UID stored in memory. If the RFID tag is authorized, the controller activates the onboard LED and enables the L298N motor driver to start the ignition motor. If an unauthorized RFID tag is detected, the motor remains OFF. The stop switch is used to stop the motor manually whenever required.

The hardware setup successfully demonstrates secure RFID authentication and intelligent ignition control for automotive security applications.

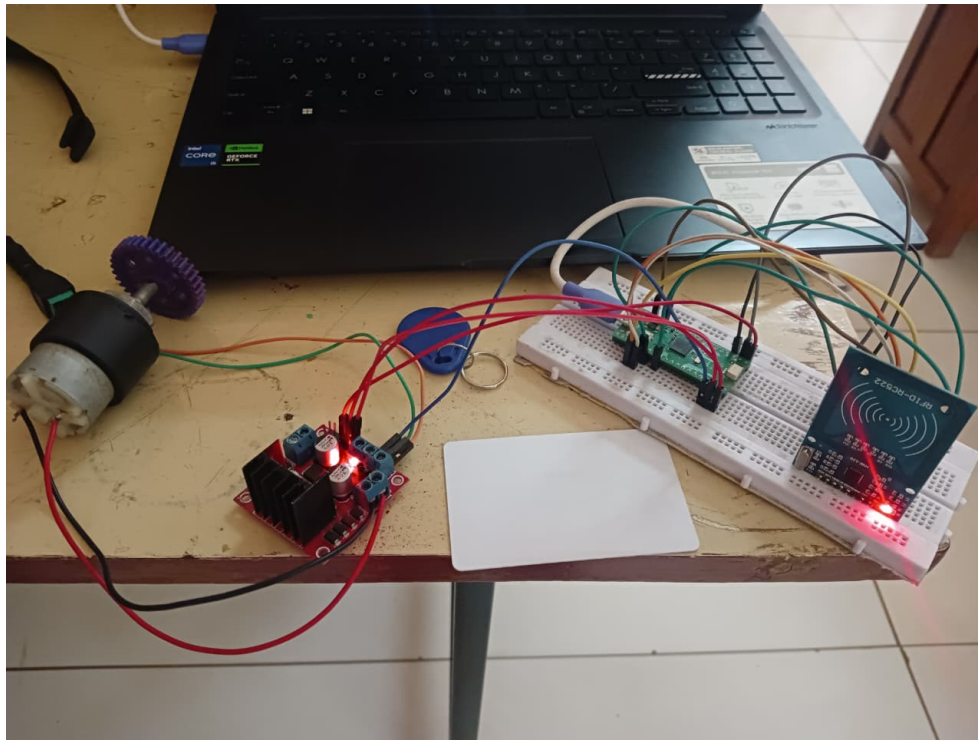


Figure 8.1: Hardware Implementation of RFID Based Smart Ignition System

Chapter 9

Code

9.1 Arduino Code

```
#include <SPI.h>
#include <MFRC522.h>

#define SS_PIN 5
#define RST_PIN 6

#define ENA 14
#define IN1 15
#define IN2 16

#define STOP_SWITCH 17

MFRC522 rfid(SS_PIN, RST_PIN);

byte authorizedUID[4] = {99, 142, 40, 3};

void setup()
{
  Serial.begin(115200);

  ///
  pinMode(LED_BUILTIN, OUTPUT);

  pinMode(ENA, OUTPUT);
  pinMode(IN1, OUTPUT);
```

```

pinMode(IN2, OUTPUT);

pinMode(STOP_SWITCH, INPUT_PULLUP);

digitalWrite(ENA, LOW);
digitalWrite(IN1, LOW);
digitalWrite(IN2, LOW);

SPI.setRX(4);
SPI.setTX(3);
SPI.setSCK(2);

SPI.begin();

rfid.PCD_Init();
'''

}

void loop()
{
if (!rfid.PICC_IsNewCardPresent())
return;

'''
if (!rfid.PICC_ReadCardSerial())
return;

bool authorized = true;

for (byte i = 0; i < 4; i++)
{
    if (rfid.uid.uidByte[i] != authorizedUID[i])
    {
        authorized = false;
        break;
    }
}
}

```

```

if (authorized)
{
    digitalWrite(LED_BUILTIN, HIGH);

    digitalWrite(ENA, HIGH);
    digitalWrite(IN1, HIGH);
    digitalWrite(IN2, LOW);

    while (digitalRead(STOP_SWITCH) == HIGH)
    {
        delay(10);
    }

    digitalWrite(ENA, LOW);
    digitalWrite(IN1, LOW);
    digitalWrite(IN2, LOW);

    digitalWrite(LED_BUILTIN, LOW);
}

else
{
    digitalWrite(ENA, LOW);
    digitalWrite(IN1, LOW);
    digitalWrite(IN2, LOW);

    digitalWrite(LED_BUILTIN, LOW);
}

rfid.PICC_HaltA();
rfid.PCD_StopCrypto1();

delay(200);
'''

}

```

Chapter 10

Applications and Future Scope

10.1 Applications

- Smart vehicle ignition systems
- Automotive security applications
- RFID-based access control systems
- Embedded automotive systems
- Intelligent vehicle authentication systems

10.2 Future Scope

- Integration with mobile application control
- Cloud-based vehicle authentication
- GPS tracking and monitoring
- Biometric-based vehicle security
- IoT-based smart vehicle systems

Chapter 11

Conclusion

The RFID Based Smart Ignition System was successfully designed, simulated, and implemented using Raspberry Pi Pico, RC522 RFID module, and L298N motor driver. The proposed system provides secure vehicle authentication using RFID technology and allows only authorized users to start the ignition motor.

The simulation results verified the proper working of the authentication and motor control logic, while the hardware implementation demonstrated reliable real-time operation. The stop switch functionality provided additional control for stopping the motor manually whenever required.

Overall, the project demonstrates a practical application of embedded systems, automotive electronics, and RFID-based security systems. The developed system is cost-effective, reliable, and suitable for modern smart vehicle security applications.

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